Test 2 is Friday. Today: every problem type from your Unit 2 quizzes.

- **Turn in Quiz 9** now.
- Test 2 has **10 questions, 20 points each**, built from the skills on Quizzes 5 to 9: about 20% Section 4.2, 20% Section 4.3, 30% polygons, 10% DMS and 20% circles.
- For each type: I work a model, then you try one. Answers appear when we check.
- Most questions have two parts, such as setup and answer. Show your work in each part.

Nothing is printed on Test 2.

Write every formula from memory, including:

- $A = P(1 + r/n)^{nt}$ and $A = Pe^{rt}$
- $\log_b x = y \iff b^y = x$
- trapezoid: $A = \frac{1}{2}(b_1 + b_2)h$; Heron: $s = (a + b + c)/2$, $A = \sqrt{(s(s-a)(s-b)(s-c))}$
- DMS: $D + M/60 + S/3600$; inscribed angle = ½ arc
- radians = degrees × $\pi/180$; $s = r\theta$; $A = \frac{1}{2}r^2\theta$
- $v = r\omega$, with ω = rpm × 2π

$A = P(1 + r/n)^{nt}$

r as a decimal · n = 2 semiannually, 4 quarterly, 12 monthly · round to the nearest cent at the end

Model: \$2,500 at 3.5% compounded monthly for 6 years

Setup: $A = 2500(1 + 0.035/12)^{12 \cdot 6} = 2500(1 + 0.035/12)^{72}$

$A \approx \$3{,}083.25$

You try: \$7,000 at 5.2% compounded quarterly for 9 years

Setup: $A = 7000(1 + 0.052/4)^{36}$

$A \approx \$11{,}143.92$

$A = Pe^{rt}$

"Compounded continuously" means use e · multiply r × t first · round to the nearest cent

Model: $1,500 at 6.2% compounded continuously for 7 years

Setup: $A = 1500e^{0.062 \cdot 7} = 1500e^{0.434}$

$A \approx \$2{,}315.13$

You try: $9,000 at 3.8% compounded continuously for 11 years

Setup: $A = 9000e^{0.418}$

$A \approx \$13{,}670.29$

Evaluate, then growth or decay

$f(t) = a \cdot b^t$ · $b > 1$: growth · $0 < b < 1$: decay

Model

$P(t) = 450(1.15)^t$, find $P(6) \to \approx 1040.88$; $1.15 > 1$, growth

$f(t) = 3000(0.92)^t$, find $f(4) \to \approx 2149.18$; $0.92 < 1$, decay

You try

$Q(t) = 1800(0.75)^t$, find $Q(3) \to \approx 759.38$; decay

$R(t) = 250(1.06)^t$, find $R(12) \to \approx 503.05$; growth

The log is the exponent

$$b^y = x \iff \log_b x = y$$

Model

(a) $5^3 = 125$ in log form $\rightarrow \log_5 125 = 3$

(b) Evaluate $\log_4 0.0625 \rightarrow 0.0625 = 1/16 = 4^{-2}$, so -2

You try

(a) $7^2 = 49$ in log form $\rightarrow \log_7 49 = 2$

(b) Evaluate $\log_2 0.0625 \rightarrow 0.0625 = 1/16 = 2^{-4}$, so -4

Take the log of both sides

$b^x = c$: $x = \ln c \div \ln b$ · $e^{x-k} = c$: $x - k = \ln c$ · four decimal places

Model: $3^x = 50$

Setup: $x \ln 3 = \ln 50$

$x = \ln 50 \div \ln 3 \approx 3.5609$

$e^{x-3} = 12 \to x - 3 = \ln 12$, $x \approx 5.4849$

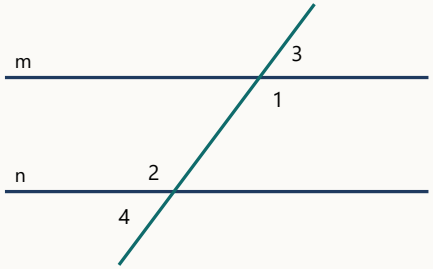
You try: $6^x = 150$

Setup: $x \ln 6 = \ln 150$

$x \approx 2.7965$

$e^{x+2} = 50 \to x = -2 + \ln 50 \approx 1.9120$

Equal or supplementary?



m ∥ n. Alternate interior (1 and 2), alternate exterior (3 and 4) and corresponding angles are **equal**. Same-side interior angles **add to 180°**.

Model: alternate interior angles (2x + 27)° and (4x − 13)°

Equation: $2x + 27 = 4x - 13 \rightarrow x = 20 \rightarrow \text{angle} = 2(20) + 27 = 67°$

You try: same-side interior angles (2x + 20)° and (4x + 10)°

$(2x + 20) + (4x + 10) = 180 \rightarrow x = 25 \rightarrow$ angles $70°$ and $110°$

Matching angles are equal; matching sides are proportional

△ABC ~ △DEF means AB ↔ DE, BC ↔ EF, AC ↔ DF. Keep one triangle on top: AB/BC = DE/EF.

Model: angle A = 38°, angle B = 75°; AB = 6, BC = 10, DE = 9

$$\text{Angle F} = \text{angle C} = 180 - 38 - 75 = 67°$$

$$\text{EF: } 6/10 = 9/x$$

$$6x = 90 \rightarrow x = 15$$

You try: angle A = 49°, angle C = 76°; AB = 14, BC = 21, DE = 12

$$\text{Angle E} = \text{angle B} = 55°$$

$$\text{EF: } 14/21 = 12/x$$

$$14x = 252 \rightarrow x = 18$$

Two areas from memory

Trapezoid: $A = \frac{1}{2}(b_1 + b_2)h$ · Heron: $s = (a + b + c)/2$, $A = \sqrt{(s(s-a)(s-b)(s-c))}$

Model: trapezoid with parallel sides 10 m and 18 m, height 7 m; triangle with sides 5, 12, 13

Trapezoid: $\frac{1}{2}(10 + 18)(7) = 98\ \text{m}^2$

$s = (5 + 12 + 13)/2 = 15$

$A = \sqrt{(15 \cdot 10 \cdot 3 \cdot 2)} = \sqrt{900} = 30$

You try: trapezoid with parallel sides 6.5 cm and 11.5 cm, height 4 cm; triangle with sides 7, 15, 20

Trapezoid: $36\ \text{cm}^2$

$s = 21$

$A = \sqrt{(21 \cdot 14 \cdot 6 \cdot 1)} = \sqrt{1764} = 42$

DMS both ways

decimal → DMS: × 60, then × 60 again (nearest second) · DMS → decimal: D + M/60 + S/3600 (four places)

Model

$54.2875° \to 0.2875 \times 60 = 17.25′$; $0.25 \times 60 = 15″ \to 54°17′15″$

$31°12′45″ \to 31 + 12/60 + 45/3600 = 31.2125°$

You try

$98.4561° \to 98°27′22″$

$17°53′24″ \to 17.8900°$

Half the arc; radians before s = rθ

central angle = its arc · inscribed angle = ½ arc · radians = degrees × π/180 · $s = r\theta$

Model: radius 8, central angle AOC = 120°

Inscribed angle ABC $= 120 \div 2 = 60°$

$120 \times \pi/180 = 2\pi/3$ rad

$s = 8(2\pi/3) = 16\pi/3 \approx 16.76$

You try: radius 10, central angle AOC = 84°

Inscribed angle $= 42°$

$84° = 7\pi/15$ rad; $s = 14\pi/3 \approx 14.66$

$A = \frac{1}{2}r^2\theta$, and $v = r\omega$

θ in radians · ω = rpm × 2π rad per minute · $v = r\omega$ in ft/min

Model: sector r = 10, $\theta = 2\pi/5$; wheel r = 2.5 ft at 160 rpm

$A = \frac{1}{2}(100)(2\pi/5) = 20\pi \approx 62.83$ square units

$\omega = 160(2\pi) = 320\pi$; $v = 2.5(320\pi) = 800\pi \approx 2513.3$ ft/min

You try: sector r = 6, $\theta = 5\pi/4$; wheel r = 1.5 ft at 210 rpm

$A = 22.5\pi \approx 70.69$ square units

$v = 630\pi \approx 1979.2$ ft/min

Where points are usually lost

- Rate as a percent (5.2) instead of a decimal (0.052).
- Wrong n: quarterly is 4, semiannually is 2, monthly is 12.
- Calling a model growth because the starting value is large. Only the base decides.
- Moving the base to the wrong side in log form.
- Rounding in the middle of a calculation.
- Solving for x but not finding the angle.
- Mismatched sides in a similar-triangle proportion.
- Skipping the semiperimeter, or a negative factor in Heron.
- Doubling an inscribed angle instead of halving the arc.
- Dividing seconds by 60 instead of 3600.
- Using degrees in $s = r\theta$ or $A = \frac{1}{2}r^2\theta$.
- Leaving ω in rpm, or using a diameter as the radius.

Before Friday

Tonight and Thursday

- Write all the formulas from memory until you can do it without looking.
- Redo every "you try" from today without notes.
- Work the Test 2 practice PDF on the course page, then check it.
- Take the Test 2 readiness check on the review page until you reach 80%.

Friday, October 30

- Test 2, in class on paper: 10 questions, 200 points, 50 minutes
- No formula sheet: everything is from memory
- Scientific calculator
- Show your work inside each question's box. A correct answer with no work gets no credit.